

Complex Analysis Prerequisites: Definitions

Complex Numbers

Definition 1 (Complex Number). A **complex number** is an expression of the form

$$z = x + iy, \quad x, y \in \mathbb{R},$$

where i is the imaginary unit satisfying $i^2 = -1$. We write $x = \operatorname{Re}(z)$ (the real part) and $y = \operatorname{Im}(z)$ (the imaginary part). The set of all complex numbers is denoted $\mathbb{C}$.

Definition 2 (Modulus and Argument). For $z = x + iy \in \mathbb{C}$, the **modulus** (absolute value) is

$$|z| = \sqrt{x^2 + y^2},$$

and the **argument** $\arg(z)$ is the angle θ such that $z = |z|(\cos \theta + i \sin \theta)$, defined up to an integer multiple of 2π .

Definition 3 (Complex Conjugate). The **conjugate** of $z = x + iy$ is $\bar{z} = x - iy$. Note that $z\bar{z} = |z|^2$.

Definition 4 (Polar and Exponential Form). Every nonzero $z \in \mathbb{C}$ can be written as

$$z = re^{i\theta}, \quad r = |z|, \quad \theta = \arg(z),$$

using Euler's formula (below).

Euler's Number and Euler's Formula

Definition 5 (Euler's Number). Euler's number e is defined as

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = \sum_{k=0}^{\infty} \frac{1}{k!} \approx 2.71828 \dots$$

[Euler's Formula] For any real number θ ,

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

Consequently, $e^{i\pi} + 1 = 0$ (Euler's identity).

Definition 6 (Complex Exponential). For $z = x + iy \in \mathbb{C}$, the **complex exponential** is defined by

$$e^z = e^{x+iy} = e^x(\cos y + i \sin y).$$

Complex Functions

Definition 7 (Complex Function). A **complex function** is a map $f : D \rightarrow \mathbb{C}$, where $D \subseteq \mathbb{C}$ is the domain. Writing $z = x + iy$ and $f(z) = u(x, y) + iv(x, y)$, we identify f with a pair of real-valued functions $u, v : D \rightarrow \mathbb{R}$ (the real and imaginary parts of f).

Definition 8 (Limit of a Complex Function). Let $f : D \rightarrow \mathbb{C}$ and let z_0 be a limit point of D . We say

$$\lim_{z \rightarrow z_0} f(z) = L$$

if for every $\varepsilon > 0$ there exists $\delta > 0$ such that $0 < |z - z_0| < \delta$ implies $|f(z) - L| < \varepsilon$.

Continuity

Definition 9 (Continuity at a Point). A function $f : D \rightarrow \mathbb{C}$ is **continuous** at $z_0 \in D$ if

$$\lim_{z \rightarrow z_0} f(z) = f(z_0).$$

f is continuous on D if it is continuous at every point of D .

Differentiability and Holomorphy

Definition 10 (Complex Differentiability). A function $f : D \rightarrow \mathbb{C}$, with D open, is **complex differentiable** at $z_0 \in D$ if the limit

$$f'(z_0) = \lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h}$$

exists, where $h \in \mathbb{C}$.

Definition 11 (Holomorphic Function). A function f is **holomorphic** (or **analytic**) on an open set $D \subseteq \mathbb{C}$ if it is complex differentiable at every point of D .

[Cauchy–Riemann Equations] If $f(z) = u(x, y) + iv(x, y)$ is complex differentiable at $z_0 = x_0 + iy_0$, then u and v satisfy

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

at (x_0, y_0) .

Convergence

Definition 12 (Convergent Sequence). A sequence (z_n) in $\mathbb{C}$ **converges** to $L \in \mathbb{C}$ if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|z_n - L| < \varepsilon$ for all $n \geq N$.

Definition 13 (Convergent Series). A series $\sum_{n=1}^{\infty} z_n$ **converges** to S if the sequence of partial sums $S_N = \sum_{n=1}^N z_n$ converges to S .

Definition 14 (Absolute Convergence). A series $\sum_{n=1}^{\infty} z_n$ **converges absolutely** if $\sum_{n=1}^{\infty} |z_n|$ converges. Absolute convergence implies convergence.

Definition 15 (Uniform Convergence). A sequence of functions $f_n : D \rightarrow \mathbb{C}$ **converges uniformly** to f on D if for every $\varepsilon > 0$ there exists N such that

$$\sup_{z \in D} |f_n(z) - f(z)| < \varepsilon \quad \text{for all } n \geq N.$$

Power Series and Region of Convergence

Definition 16 (Power Series). A **power series** centered at z_0 is an expression

$$\sum_{n=0}^{\infty} a_n (z - z_0)^n, \quad a_n \in \mathbb{C}.$$

Definition 17 (Radius of Convergence). The **radius of convergence** R of a power series is given by

$$\frac{1}{R} = \limsup_{n \rightarrow \infty} |a_n|^{1/n}.$$

The series converges absolutely for $|z - z_0| < R$ and diverges for $|z - z_0| > R$.

Analytic Continuation

Definition 18 (Analytic Continuation). Let f_1 be holomorphic on a domain $D_1 \subseteq \mathbb{C}$. If f_2 is holomorphic on a domain D_2 with $D_1 \cap D_2 \neq \emptyset$, and $f_1 = f_2$ on $D_1 \cap D_2$, then f_2 is called an **analytic continuation** of f_1 to D_2 .

[Identity Theorem] If f and g are holomorphic on a connected open set D and $f = g$ on a subset of D having a limit point in D , then $f = g$ on all of D . Consequently, an analytic continuation, when it exists, is **unique**.

Meromorphic Functions and Poles

Definition 19 (Meromorphic Function). A function f is **meromorphic** on a domain D if it is holomorphic on D except for a set of isolated points, at each of which f has a pole.

Definition 20 (Pole). A point z_0 is a **pole** of f of order m if f can be written near z_0 as

$$f(z) = \frac{g(z)}{(z - z_0)^m},$$

where g is holomorphic near z_0 and $g(z_0) \neq 0$. A pole of order 1 is called a **simple pole**.

The Gamma Function

Definition 21 (Gamma Function). For $\operatorname{Re}(s) > 0$, the **Gamma function** is defined by

$$\Gamma(s) = \int_0^\infty t^{s-1} e^{-t} dt.$$

It extends to a meromorphic function on $\mathbb{C}$ with simple poles at $s = 0, -1, -2, \dots$, and satisfies $\Gamma(n+1) = n!$ for $n \in \mathbb{N}$.

The Riemann Zeta Function

Definition 22 (Riemann Zeta Function). For $\operatorname{Re}(s) > 1$, the **Riemann zeta function** is defined by

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

[Euler Product Formula] For $\operatorname{Re}(s) > 1$,

$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}.$$

Taylor Series, Maclaurin Series, and Singularities

Taylor Series

[Taylor's Theorem for Holomorphic Functions] Let f be holomorphic on an open disk $D = \{z : |z - z_0| < R\}$. Then f has a convergent power series representation

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n, \quad |z - z_0| < R,$$

called the **Taylor series** of f centered at z_0 , where the coefficients are given by

$$a_n = \frac{f^{(n)}(z_0)}{n!} = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta,$$

with γ any positively oriented simple closed contour in D around z_0 .

A key fact of complex analysis: every holomorphic function is automatically **analytic**, meaning it is locally equal to its own convergent Taylor series. This is stronger than the real case, where infinite differentiability does not guarantee a convergent Taylor expansion.

Definition 23 (Taylor Series — General Form). The **Taylor series** of f about a point z_0 is

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \frac{f''(z_0)}{2!}(z - z_0)^2 + \frac{f'''(z_0)}{3!}(z - z_0)^3 + \dots$$

Maclaurin Series

Definition 24 (Maclaurin Series). A **Maclaurin series** is a Taylor series centered at $z_0 = 0$:

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} z^n = f(0) + f'(0)z + \frac{f''(0)}{2!} z^2 + \dots$$

[Common Maclaurin Series]

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}, \quad |z| < \infty$$

$$\sin z = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n+1}}{(2n+1)!}, \quad |z| < \infty$$

$$\cos z = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n}}{(2n)!}, \quad |z| < \infty$$

$$\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n, \quad |z| < 1$$

$$\log(1+z) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} z^n, \quad |z| < 1$$

Laurent Series

Definition 25 (Annulus). An **annulus** centered at z_0 is a region of the form

$$A = \{z \in \mathbb{C} : r < |z - z_0| < R\}, \quad 0 \leq r < R \leq \infty.$$

[Laurent's Theorem] Let f be holomorphic on an annulus $A = \{z : r < |z - z_0| < R\}$. Then f has a unique series representation

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$

convergent on A , called the **Laurent series** of f , where

$$a_n = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta,$$

with γ a positively oriented circle in A centered at z_0 . The part $\sum_{n=0}^{\infty} a_n (z - z_0)^n$ is the **analytic (regular) part**, and $\sum_{n=1}^{\infty} a_{-n} (z - z_0)^{-n}$ is the **principal part**.

Isolated Singularities

Definition 26 (Isolated Singularity). A point z_0 is an **isolated singularity** of f if f is holomorphic on some punctured disk $0 < |z - z_0| < R$ but not at z_0 itself.

Given an isolated singularity, f has a Laurent expansion

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n$$

on a punctured neighborhood of z_0 . Singularities are classified according to the principal part $\sum_{n=1}^{\infty} a_{-n}(z - z_0)^{-n}$.

Definition 27 (Removable Singularity). z_0 is a **removable singularity** of f if $a_n = 0$ for all $n < 0$, i.e., the principal part vanishes entirely. Equivalently, $\lim_{z \rightarrow z_0} f(z)$ exists (finitely), and f can be redefined at z_0 to be holomorphic there.

[Riemann's Removable Singularity Theorem] An isolated singularity z_0 of f is removable if and only if f is bounded on some punctured neighborhood of z_0 .

Definition 28 (Pole). z_0 is a **pole** of order $m \geq 1$ if $a_{-m} \neq 0$ and $a_n = 0$ for all $n < -m$. That is, the Laurent series has the form

$$f(z) = \frac{a_{-m}}{(z - z_0)^m} + \cdots + \frac{a_{-1}}{z - z_0} + \sum_{n=0}^{\infty} a_n(z - z_0)^n.$$

A pole of order 1 is a **simple pole**; order 2 is a **double pole**, etc. Equivalently, z_0 is a pole of order m if and only if

$$\lim_{z \rightarrow z_0} (z - z_0)^m f(z)$$

exists and is nonzero and finite.

z_0 is a pole of f if and only if $\lim_{z \rightarrow z_0} |f(z)| = \infty$.

Definition 29 (Essential Singularity). z_0 is an **essential singularity** of f if $a_n \neq 0$ for infinitely many negative n , i.e., the principal part has infinitely many nonzero terms.

[Casorati–Weierstrass Theorem] If z_0 is an essential singularity of f , then for every punctured neighborhood U of z_0 , the image $f(U)$ is dense in $\mathbb{C}$.

[Picard's Great Theorem] If z_0 is an essential singularity of f , then on every punctured neighborhood of z_0 , f takes every complex value, with at most one exception, infinitely often.

$$\text{Removable: } f(z) = \frac{\sin z}{z} \text{ at } z_0 = 0, \quad \text{since } \frac{\sin z}{z} = 1 - \frac{z^2}{3!} + \frac{z^4}{5!} - \cdots$$

Simple pole: $f(z) = \frac{1}{z}$ at $z_0 = 0$

Pole of order 3: $f(z) = \frac{1}{z^3}$ at $z_0 = 0$

Essential singularity: $f(z) = e^{1/z}$ at $z_0 = 0$, since $e^{1/z} = \sum_{n=0}^{\infty} \frac{1}{n!} z^{-n}$

Singularities at Infinity

Definition 30 (Singularity at Infinity). *The behavior of f at $z = \infty$ is classified by studying $g(w) = f(1/w)$ at $w = 0$. The type of singularity of g at $w = 0$ (removable, pole of order m , or essential) determines the corresponding type for f at ∞ .*

Poles and the Zeta Function

In the context of $\zeta(s)$: the Dirichlet series $\sum n^{-s}$ has a **simple pole** at $s = 1$ with residue 1, i.e.

$$\zeta(s) = \frac{1}{s-1} + \gamma + O(s-1) \quad \text{as } s \rightarrow 1,$$

where γ is the Euler–Mascheroni constant. This is the unique singularity of $\zeta(s)$ on all of $\mathbb{C}$ after analytic continuation, making ζ meromorphic on $\mathbb{C}$ with exactly one pole.